

Samuel Granados

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Education

Instituto Tecnológico y de Estudios Superiores de Monterrey

Monterrey, N.L.

B.S. in Electronics Engineering

Expected Graduation: December 2027

- GPA 3.8/4.0 | Full Academic Leadership Scholarship (100%)
- Specialization: Power Electronics & Hardware Design

Hong Kong University of Science and Technology (HKUST)

Hong Kong

Exchange Program – Coursework: IC Design, FPGA, Transnational Legal Issues & Dispute Settlement

Fall 2026

- Started early-stage power electronics research in Prof. Wang Ping's lab.

Work Experience

Carrier

Santa Catarina, N.L.

Electronics Validation & Automation Intern | Electrical CoE

June 2025 – March 2026

- **Multiplexer:** Designed an RS-485 protocol multiplexer (ATmega328 + DG409) on a custom Altium PCB; built a C# GUI to evaluate real-time data from boards sharing the same address.
- **RCA & Validation:** Conducted Root Cause Analysis on PCB failures unresolved for over a year, proposing fixes; ran component validation testing (environmental chambers, HIL rigs) for thermal shocks, deploying results via a C# interface into Power BI.

Independent Test & Validation Intern | Software & Controls CoE

March 2026 – Present

- **Small Sim:** Designed an Altium shield for an STM32H563ZI running FreeRTOS, replicating residential 24VAC/PWM sensor inputs to validate embedded firmware.

Co-curricular Activities & Achievements

RoBorregos

Monterrey, N.L.

Director of Finance & Electronics PM, @HOME League

June 2025 – Present

- As Director of Finance, secured \$400K+ MXN in sponsorships managing a \$700K+ MXN budget for national/international competitions; as Electronics PM, led the BMS/power distribution design achieving **1st Place Nationally** (April 2026 – see Engineering Projects).

PLEI Asia Mission

International

Selected Mission Member

June – August 2026

- First fully sponsored student selected for this international business mission; sourced 30+ potential suppliers for DYL Medic (6 visited on-site) and studied a leading nearshoring agency's Philippines market model for Intugo.

Engineering Projects

Power Electronics

- **Variable Frequency Drive (VFD) – AMI Automation:** Designed and implemented a two-level Voltage Source Inverter (VSI) driving a 3-phase induction motor via scalar V/f control; developed sinusoidal PWM (S-PWM) firmware in C (STM32, HAL) generating three 120°-phased references at a 100µs sample rate, validated across a 9–36Hz operating range with inverter output up to 109 V_{RMS}.
- **Hybrid PV+BESS Pumping Microgrid (Sonora):** Sized a 42.93kWp PV array and 91.98kWh LiFePO4 BESS with VFD/MPPT control for a 25HP agricultural pump; modeled techno-economic viability (6.75-yr payback, 14.2% IRR).
- **Bendix:** Designed and simulated a custom inductor using COMSOL Multiphysics, optimizing magnetic performance and validating design constraints in collaboration with industry engineers.

Embedded Systems, DSP & IoT

- **Visual-Inertial Odometry Rover (DSP):** Built a dual-core ESP32-S3 2D positioning system fusing IMU quaternions with SAD-based optical flow via an Extended Kalman Filter and ZUPT drift correction; achieved <20ms ESP-NOW latency and ~5% position error on a lunar-rover-inspired use case.
- **AgroNode (IoT):** Co-designed a multi-protocol IoT node (custom PCB, ESP32-S3) with ESP-NOW, LoRa, and 2G cellular; diagnosed a LoRa antenna mismatch via Friis/reflection-coefficient analysis, quantifying a 20dB loss behind a 3.2km→1m range gap.
- **OPmobility Challenge 2025:** Designed and validated an ECU diagnostic prototype for automotive LED lighting on an ATmega328P; implemented ADC-based fault detection (overvoltage, undervoltage, open/short, overtemperature), full-duplex UART reporting, and persistent EEPROM logging, verified via unit/integration testing.

Skills and Interests

Programming & Control: C++, C#, C, Python, MATLAB, Simulink.

Power Electronics: Voltage Source Inverters, Scalar V/f & S-PWM Motor Control, Power Converters, Magnetics Design (COMSOL).

Embedded Systems: Microcontrollers (ESP32, AVR/ATmega, PIC18, TI F28379D, STM32), Sensor Fusion (EKF), DSP, Data Acquisition (DAQ) Systems, Verilog, FPGA, IC Design.

Communication Protocols: CAN, RS-485, I2C, UART, TCP/IP, SCPI, LoRa, ESP-NOW, 2G/GPRS.

Design: Hardware design (Altium, KiCad, OrCad), Simulation (LTspice, Multisim).

Electronics: Analog IC Design, ADC/DAC Principles, Semiconductor Devices, Optoelectronics, Amplifiers.

Languages: English (B2 - 6.5 IELTS), Spanish (Native).

Certifications: Google Data Analytics (2025), Lean Six Sigma Green Belt (2024), MIT NanoLab (2024).